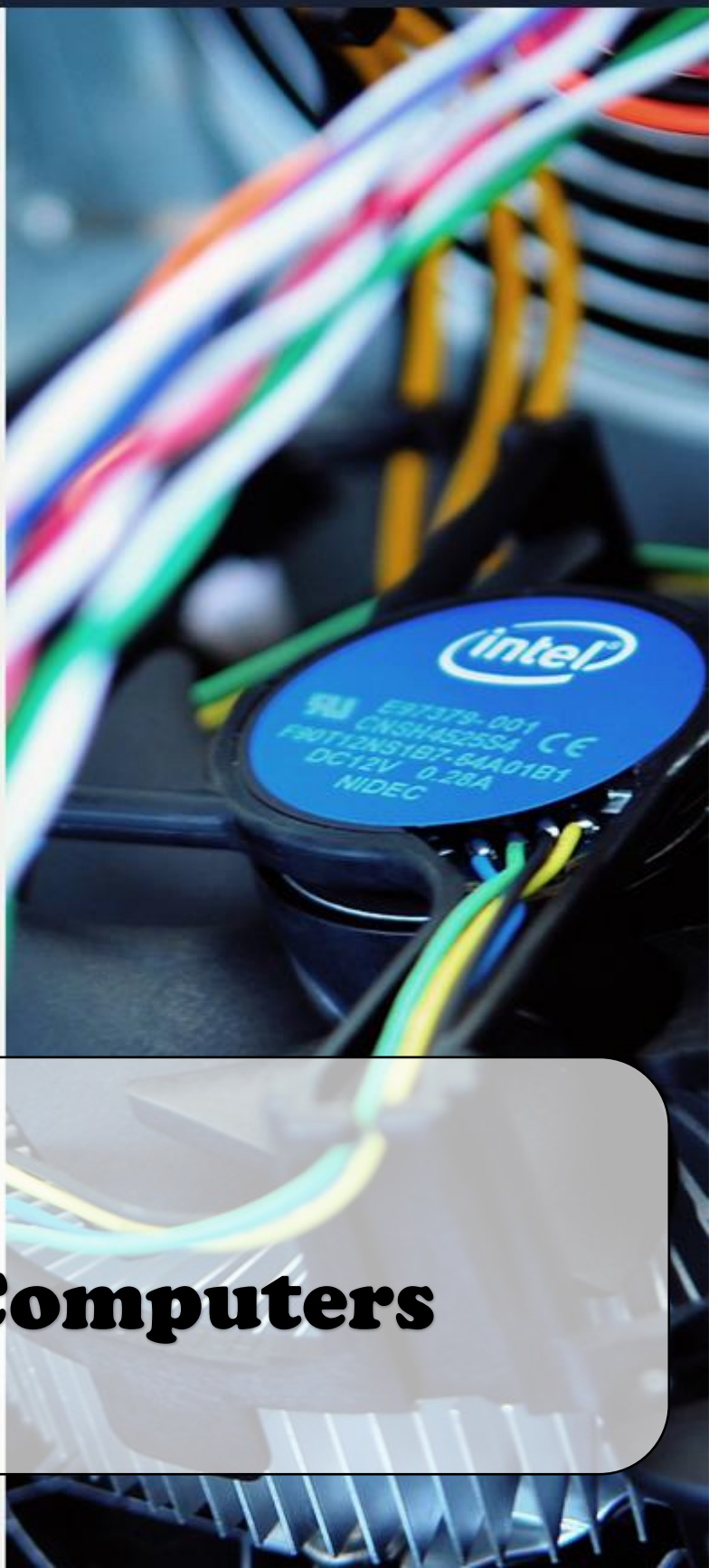
GOVERNMENT PROPERTY
NOT FOR SALE

DepED
DEPARTMENT OF EDUCATION

COMPUTER SYSTEMS SERVICING NC II
GRADE 11

QUARTER 1
COC 1:
Install and Configure Computer Systems



Module 1:

Types of Computers

Computer System Servicing – Grade 11
Quarter 1 – Module 1: Types of Computer
First Edition, 2020

Republic Act 8293, section 176 states that: No copyright shall subsist in any work of the Government of the Philippines. However, prior approval of the government agency or office wherein the work is created shall be necessary for exploitation of such work for profit. Such agency or office may, among other things, impose as a condition the payment of royalties.

Borrowed materials (i.e., songs, stories, poems, pictures, photos, brand names, trademarks, etc.) included in this module are owned by their respective copyright holders. Every effort has been exerted to locate and seek permission to use these materials from their respective copyright owners. The publisher and authors do not represent nor claim ownership over them.

Published by the Department of Education Division of Pasig City

Development Team of the Module

Writer: Aiza T. Tuplano

Editors: Name

Reviewers: Felix C. Vergara, Jr.

Illustrator: Name

Layout Artist: Name

Management Team: Ma. Evalou Concepcion A. Agustin
OIC-Schools Division Superintendent

Dr. Aurelio G. Alfonso
OIC-Assistant Schools Division Superintendent

Dr. Victor Javena
OIC – Chief Curriculum Implementation Division

Wilma Q. Del Rosario, Division EPS In Charge of LRMDS
Librada L. Agon, Education Program Specialist (EPP-TLE-TVL)

Printed in the Philippines by _____

Department of Education – Division of Pasig City

Office Address: Caruncho Avenue, San Nicolas, Pasig City
Telefax:
E-mail Addres

Introductory Message

For the facilitator:

Welcome to the Computer Systemss Servicing COC1: Install and Configure Computer System Module 1 on Types of Computers!

This module was collaboratively designed, developed, and reviewed by educators from Schools Division Office of Pasig City headed by its Officer-In-Charge Schools Division Superintendent, Ma. Evalou Concepcion A. Agustin in partnership with the Local Government of Pasig through its mayor, Honorable Vico Sotto. The writers utilized the standards set by the K to 12 Curriculum using the Most Essential Learning Competencies (MELC) while overcoming their personal, social, and economic constraints in schooling.

This learning material hopes to engage the learners into guided and independent learning activities at their own pace and time. Further, this also aims to help learners acquire the needed 21st century skills especially the 5 Cs namely: Communication, Collaboration, Creativity, Critical Thinking and Character while taking into consideration their needs and circumstances.

In addition to the material in the main text, you will also see this box in the body of the module:

Notes to the Teacher

This contains helpful tips or strategies that will help you in guiding the learners.

As a facilitator you are expected to orient the learners on how to use this module. You also need to keep track of the learners' progress while allowing them to manage their own learning. Moreover, you are expected to encourage and assist the learners as they do the tasks included in the module.

For the learner:

Welcome to the Computer Systems Servicing COC1: Install and Configure Computer System Module 1 on Types of Computers!

The hand is one of the most symbolized part of the human body. It is often used to depict skill, action, and purpose. Through our hands we may learn, create, and accomplish. Hence, the hand in this learning resource signifies that you as a learner is capable and empowered to successfully achieve the relevant competencies and skills at your own pace and time. Your academic success lies in your own hands!

This module was designed to provide you with fun and meaningful opportunities for guided and independent learning at your own pace and time. You will be enabled to process the contents of the learning material while being an active learner.

This module has the following parts and corresponding icons:

Expectation - These are what you will be able to know after completing the lessons in the module

Pre-test - This will measure your prior knowledge and the concepts to be mastered throughout the lesson.

Recap - This section will measure what learnings and skills that you understand from the previous lesson.

Lesson- This section will discuss the topic for this module.

Activities - This is a set of activities you will perform.

Wrap Up- This section summarizes the concepts and applications of the lessons.

Valuing-this part will check the integration of values in the learning competency.

Post-test - This will measure how much you have learned from the entire module. Ito po ang parts ng module

EXPECTATION

At the end of this module, student will be able to:

1. identify the types of computers;
2. classify the computers according to its function and uses;
3. appreciate the benefits of computers in their daily lives.

PRE – TEST

MULTIPLE CHOICE:

Direction: Read the following statement correctly and box out the correct answer.

1. What type of computer is being used for Space Explorations?
   a. Micro Computer b. Supercomputer
   c. Mainframe computer d. Minicomputer

2. Which of the following is an example of Micro Computer?
   a. Tianhe-2A b. Fujitsu ICL VME
   c. Personal Digital Assistant d IBM Midrange Computer

3. Which of the following type of computer has an 8 to 18 bits word size?
   a. Micro Computer b. Supercomputer
   c. Mainframe computer d. Minicomputer

4. This is also called as the "fastest type of computer"
   a. Micro Computer b. Supercomputer
   c. Mainframe computer d. Minicomputer

5. Which of the following type of computer use microprocessor?
   a. Micro Computer b. Supercomputer
   c. Mainframe computer d. Minicomputer

6. What type of computer that when it comes to its speed and storage is mostly used by Government organizations?
   a. Micro Computer b. Supercomputer
   c. Mainframe computer d. Mini Computer

7. This type of computer is often called “Midrange Computer”
   a. Micro Computer b. Supercomputer
   c. Mainframe computer d. Minicomputer

8. Personal Computer, smartphones and Calculator are type of what?
   a. Micro Computer b. Supercomputer
   c. Mainframe computer d. Minicomputer

9. Minicomputer is mostly used by what?
   a. Research Organization b. Government Organization
   c. Personal/Individual Use d. Business Organization

10. Which of the following does not belong to Supercomputer use?
   a. Space Exploration b. Nuclear Power Testing
   c. Earthquake Studies d. production Process

RECAP

Where do I belong?

Directions:
1. Box out the word/s that belong to Occupational Health and Safety (OHS)
2. Encircle the word/s that belongs to Tools, Material and Equipment

Wash hands thoroughly	Hazard	UTP Cable
Electromagnetic Strap	Gas Range	Spatula
Wear PPE	Remove accessories	Risk
Flour	Screw drivers	Sketch Pad
Tester	5S	Use gloves

LESSON

MODULE 1: TYPES OF COMPUTERS

I. Super Computer

The Supercomputers are the most powerful computers regarding performance and data processing. There are specialized computers used by large corporations, and task specific computers.

They can do millions of math operations in one second. They 're also called the "math crunchers". Supercomputer has thousands of processors, hundreds of gigabytes of RAM, and hard drives that allow hundreds of gigabytes of storage space, and sometimes thousands.

Where it is being used?

- ✓ For Research
- ✓ For Space Exploration
- ✓ For Earthquake Studies
- ✓ For Weather Forecasting
- ✓ For Nuclear Power Testing
- ✓ For Government Systems

II. Mainframe Computers

Mainframes are not as powerful as supercomputers, they are certainly very expensive nevertheless, and many large companies and governmental organizations are using mainframes to operate their company. Because of its scale, the Mainframe computers can be lodged in large, air-conditioned spaces. Nowadays a Mainframe is a very large and expensive computer capable of supporting hundreds, or even thousands, of users simultaneously.

IBM Mainframes: IBM z Systems z13 (left) and IBM LinuxONE Rockhopper. (right)

Where it is being used?

- ✓ For Telephone Companies for connecting telephone calls
- ✓ For TV Network to send shows to million of homes
- ✓ For Transportation Department used for traffic signals, railroad and subway
- ✓ For Bank that keeps track of money flow on computer
- ✓ For Supermarkets to keep track of their products and prices
- ✓ For huge companies or employers that organize their worker's record
- ✓ For Electric company that keep track on their customers electric consumption, bills and payment.

II. Mini Computers

Minicomputers are often referred to as Midrange Machines. These are small machines and can be accommodated as super-computers and mainframes on a disc with not as processing and data storage capabilities.

Minicomputers generally have a word size of 8–18 bits (register size); a memory size of 32,000–64,000 16-bit words, or 16,000–32,000 32-bit words; a processing speed of 200–300 kilo-instructions per second (KIPS); and a price in the range of $3,000 to $50,000.

The mini was relatively small, less than 0.6-meters (two-feet) wide and mounted in a rack. Its advent also spurred the development of peripherals that had to be developed at a price consistent with that of the mini while providing satisfactory performance. delicate systems.

Data General Nova, serial number 1, the first 16-bit minicomputer, on display at the Computer History Museum

Where it is being used?	**Popular Mini Computers**
✓ Large Company ✓ An Organization ✓ Business ✓ Production type of Company	✓ K-202 ✓ Texas Instrument TI-990 ✓ SDS-92 ✓ IBM Midrange computers

II. Micro Computer

A computer system that utilizes a microprocessor as its central control and arithmetic element. The personal computer is one form. The power and price of a microcomputer is determined partly by the speed and power of the processor and partly by the characteristics of other components of the system, i.e. the memory, the disk units, the display, the keyboard, the flexibility of the hardware, and the operating system and other software.

The Micro-Computers are specially designed for general usage like entertainment, education, and work purposes. Desktop Computers or Personal Computer, Gaming consoles, Sounds and Navigation system of a car, Netbook, Notebooks, PDA, Tablet PC's, Smartphones, Calculators are all type of Micro-computers.

Some examples of Micro Computers

Where it is being used?	**Popular Mini Computers**
✓ General Uses ✓ Personal Uses	✓ Apple ✓ Dell ✓ Samsung ✓ Sony ✓ Toshiba

ACTIVITIES

ACTIVITY#1: **PUT ME IN!**

Direction: Group each word/phrase by writing them with their corresponding category.

Super Computer	Mainframe Computer	MiniComputer	Micro Computer

Smartphones	Math Crunchers	200-300kips processing speed
Tablet	Space Exploration	Has hundred gigabytes of RAM
PIZ DAINT	Personal Computer	0.6m wide and mounted in rack
Summit	Use for TV Networks	Thousands of microprocessors
PDA	IBM z System z13	IBM Midrange Computer
Hitachi Z800	Fujitsu ICL VME	use by Telephone company
Sunway	Weather Forecasting	use by Electric Company
use by Supermarket	Data General Nova	

WRAP – UP

A computer is a machine programmable to manipulate the symbols. Its key features are, it responds in a well-defined manner to a particular set of instructions, it can execute a pre-registered list of instructions (a program), it can quickly store and retrieve large quantities of data.

Computers can usually be classified by size and power, such type of computers are: Supercomputers, which has a thousand of microprocessors and hundreds of gigabytes of Ram, which is used for massive type of data such as in Research and Discovery, Space Exploration and Weather Forecasting. Mainframe Computers is next to Supercomputer, this is the second type of computer in terms of size and power. Mainframes can support many of hundreds, or even thousands, of users at the same time.

Minicomputer on the other hand, is a multi-user computer capable of supporting up to hundreds of users simultaneously and used by Bank Organization. Lastly is Micro Computer, which is powered by microprocessors for general use such as entertainment education and work purposes. And this type of computer is currently the most use of individuals.

VALUING

3 -2 -1 Approach

3 Things that you've learned today	2 lesson that you can apply on your daily life

1 thing that you want to know/explore more

POST TEST

MULTIPLE CHOICE:

Direction: Read the following statement carefully and box out of the correct answer.

1. What type of computer is being used for Space Explorations?
   a. Micro Computer b. Supercomputer
   c. Mainframe computer d. Minicomputer

2. Which of the following is an example of Micro Computer?
   a. Tianhe-2A b. Fujitsu ICL VME
   c. Personal Digital Assistant d. IBM Midrange Computer

3. Which of the following type of computer has an 8 to 18 bits word size?
   a. Micro Computer b. Supercomputer
   c. Mainframe computer d. Minicomputer

4. This is also called as the "fastest type of computer"
   a. Micro Computer b. Supercomputer
   c. Mainframe computer d. Minicomputer

5. Which of the following type of computer use microprocessor?
   a. Micro Computer b. Supercomputer
   c. Mainframe computer d. Minicomputer

6. What type of computer that when it comes to its speed and storage is mostly used by Government organizations?
   a. Micro Computer b. Supercomputer
   c. Mainframe computer d. Minicomputer

7. This type of computer is often called "Midrange Computer"
   a. Micro Computer b. Supercomputer
   c. Mainframe computer d. Minicomputer

8. Personal Computer, smartphones and Calculator are type of what?
   a. Micro Computer b. Supercomputer
   c. Mainframe computer d. Minicomputer

9. Minicomputer is mostly used by what?
   a. Research Organization b. Government Organization
   c. Personal/Individual Use d. Business Organization

10. Which of the following does not belong to Supercomputer use?
   a. Space Exploration b. Nuclear Power Testing
   c. Earthquake Studies d. production Process

Super Computer	Mainframe Computer	Mini Computer	Micro Computer
Math Crunchers	Use for TV Networks	200-300kips processing speed	Smartphones
Has hundred gigabytes of RAM	Thousand of microprocessors	0.6m wide and mounted in rack	Tablet
PIZ DAINT	Space Exploration	K-202	Personal Computer
Summit	Hitachi Z800	IBM Midrange Computer	PDA
Weather Forecasting	use by Telephone company	IBM z System z13	
Sunway	use by Electric Company	use by Supermarket	
	Fujitsu ICL VME	Data General Nova	

ACTIVITY1: PUT ME IN

1. B	2. B	3. D	4. B	5. A
6. C	7. A	8. A	9. D	10. D

PRE/POST TEST

KEY TO CORRECTION

R E F E R E N CE S

*, Name. “Four Types of Computers - Byte-Notes,” October 2, 2019.

https://byte-notes.com/four-types-computers/.

*, Name. “Four Types of Computers - Byte-Notes.” Byte, October 2, 2019.

https://byte-notes.com/four-types-computers/.

. “Karin, Sid; Bruch, Kimberly Mann.” .". Encyclopedia.com, June 17, 2020.

https://www.encyclopedia.com/computing/news-wires-white-papers-and-books/supercomputers.

“Minicomputer.” Wikipedia. Wikimedia Foundation, February 23, 2020.

https://en.wikipedia.org/wiki/Minicomputer.

“Minicomputers.” Computer Sciences. Encyclopedia.com, June 8, 2020.

https://www.encyclopedia.com/computing/news-wires-white-papers-and-books/minicomputers.

“Types of Computer.” Accessed June 17, 2020.

https://www.cs.cmu.edu/~fgandon/lecture/uk1999/computers_types/. Accessed June 17, 2020.
https://www.cs.cmu.edu/~fgandon/lecture/uk1999/computers_types/.

Daintith, John. “Microcomputer.” .". Encyclopedia.com, June 17, 2020.

https://www.encyclopedia.com/science-and-technology/computers-and-electrical-engineering/computers-and-computing/microcomputer.

Gray, Ellen. “NASA Supercomputers Power COVID-19 Research.” NASA.

NASA, May 28, 2020. https://www.nasa.gov/feature/esd/2020/nasa-supercomputers-power-covid-19-research.
https://en.wikipedia.org/wiki/Mainframe_computer.

Time, and Ventana al Conocimiento. “The Top 10 Supercomputers, the New

Scientific Giants.” OpenMind, October 1, 2019.
https://www.bbvaopenmind.com/en/technology/innovation/the-top-10-supercomputers-the-new-scientific-giants/.